ETC144/ETC244

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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END AND BACKLOG SUBJECT EXAMINATIONS – SEPTEMBER /OCTOBER 2023

Program	:	**B.E. – Common to all Programs**	**Semester**	:	**I / II**
Course Name	:	**Renewable Energy Sources**	**Max. Marks**	:	**100**
Course Code	:	**ETC144 / ETC244**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Briefly explain the principles of Renewable energy.	CO1	(10)
	b)	List the advantages and draw backs of Renewable energy.	CO1	(10)
2.	a)	What is IOE and what are the benefits of IOE?	CO1	(06)
	b)	List and briefly explain the applications of Renewable energy.	CO1	(10)
	c)	What are the main four types of sustainable energy?	CO1	(04)

UNIT – II

3.	a)	With a neat sketch explain pyranometer.	CO2	(10)
	b)	Explain solar distillation with a diagram.	CO2	(10)
4.	a)	What are the advantages and applications of solar photovoltaic system?	CO2	(10)
	b)	Explain the principle of a solar cell.	CO2	(06)
	c)	Define beam and diffused radiation.	CO2	(04)

UNIT – III

5.	a)	What are the major problems associated with wind energy.	CO3	(10)
	b)	With a sketch explain Horizontal axis wind turbine.	CO3	(10)
6.	a)	Sketch and explain Downdraft gasifier.	CO3	(10)
	b)	List the advantages and disadvantages of biomass energy.	CO3	(10)

UNIT- IV

7.	a)	Discuss the advantages and disadvantages of tidal energy.	CO4	(06)
	b)	What is tidal energy? Explain the working of a tidal power plant with a neat sketch.	CO4	(08)
	c)	Discuss the problems associated with geothermal energy conversion system.	CO4	(06)
8.	a)	Discuss the theory and working principle of ocean thermal energy conversion systems.	CO4	(08)
	b)	Explain with neat diagram the working of a geothermal power plant.	CO4	(06)
	c)	Discuss the problems associated with ocean thermal energy conversion system.	CO4	(06)

UNIT – V

9.	a)	Draw a line diagram and explain the working of a hydrogen fuel cell.	CO5	(10)
	b)	Brief about different types of fuel cells.	CO5	(10)
10	a)	Discuss the various methods of production of hydrogen for use as an energy carrier. What are the various methods of hydrogen storage?	CO5	(10)
	b)	Draw the line diagram which shows the concept of zero energy by explaining in brief.	CO5	(10)

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